

Milestone 1 Performance & EDA Report

Competition:Smart MCQ Solver Challenge

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Focus Area:Exploratory Data Analysis, Text Preprocessing, Baseline Evaluation

1. Exploratory Data Analysis & Label Distribution

A rigorous assessment of target label class distribution was performed across the 2,000 training records in `train.csv`. Understanding class balance is highly critical for natural language processing and classification pipelines to recognize and correct for inherent dataset biases.

Answer Option	Occurrences / Frequency Count	Percentage Distribution
B (Most Frequent)	490	24.50%
C	459	22.95%
A	369	18.45%
D	358	17.90%
E (Least Frequent)	324	16.20%

Sum of Most Frequent and Least Frequent Options (B + E)

814

2. Text Preprocessing & Vocabulary Size

To eliminate superficial linguistic variations caused by punctuation and casing, standard text normalization workflows were established for the `prompt` column:

- Conversion of all string elements to lowercase.
- Removal of standard punctuation via `string.punctuation` translation matrix.
- Tokenization based on whitespace splits.

Cleaned Prompt Vocabulary Size

859 unique words

Row ID 1 Remaining Tokens (Post Stop-Words)

13 words

Filtered using

`sklearn.feature_extraction.text.ENGLISH_STOP_WORDS`.

3. TF-IDF Feature Extraction & Cosine Similarity

A global feature space was constructed by horizontally combining all text blocks (`prompt + A + B + C + D + E`) across each observation. A default `TfidfVectorizer(stop_words='english')` was fitted onto the unified text corpus to compute weights.

Metric / Configuration Element	Verified Value / Output
Total Generated TF-IDF Feature Columns (Vocabulary Size)	2,807
Row ID 1 Prompt & Option A Cosine Similarity Score	0.1847
Full Dataset Match Accuracy (Highest Cosine Similarity)	13.05%

Analysis Note: The maximum-similarity match accuracy of 13.05% falls below random chance (20%). This indicates a strong presence of Lexical Distractors—incorrect choices purposefully designed to mimic the vocabulary of the question prompt to mislead simple keyword-matching frameworks.

4. Baseline Framework Performance (MAP@3 Summary)

The primary benchmark evaluation score chosen for this pipeline is **Mean Average Precision at 3 (MAP@3)**. The metric computes precision penalties according to the specific position of the accurate answer in the top three predictions:

$$MAP@3 = \frac{1}{\text{Rank of Correct Answer}}$$

Pipeline / Evaluation Strategy	Top 3 Prediction Order Structure	Overall MAP@3 Score
Majority Class Baseline	Static prediction list: <code>[B, C, A]</code> for all rows	0.4212
TF-IDF Cosine Pipeline	Dynamic ranking ordered by similarity score	0.2962

5. Strategic Recommendations

- **Transition to Dense Embeddings:** Keyword matching fails due to surface-level distractors. Utilizing semantic encoders (e.g., Sentence-Transformers) is required.
- **Cross-Validation Setup:** Implement a 5-fold Stratified Cross-Validation setup to prevent model overfitting.